

When does cortical coherence carry shared content?

Compatibility and self-reconstruction as testable conditions

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Abstract

Synchrony accounts treat collective phase order as the mechanism that integrates distributed cortical activity. We show formally that phase order alone establishes neither shared content nor a self-model, and state what additionally does. We separate three conditions: coherence (collective phase order), compatibility (agreement of independently decoded local descriptions where regions overlap) and self-reconstruction (accurate recovery of the global state from an internal encoding). Compatible local descriptions determine a unique global state. Coherence bounds disagreement between overlapping descriptions only through shared Lipschitz encoders, and the bound grows as the square root of the population read, so on a large population it constrains nothing. A contracting self-map fixes one state but reconstructs accurately only states within a diameter set by its error and contraction factor; a broad family needs input-conditioned maps and discriminating codes. Lean 4 verifies these implications, and one model satisfies all eight connecting hypotheses together; this shows

consistency, not cortical realization. The conditions yield a disconfirmable comparison: decoders fitted independently on overlapping regions should agree more on reported than on matched unreported stimuli at equal phase coherence. An exploratory EEG analysis is a measurement check only; its pooled estimator places a perfectly synchronized signal at zero coherence. Resource and implementation constraints, and a held-out onset comparison during anaesthetic emergence, are stated briefly. Unity and a minimal self remain interpretations that need an independent experiential endpoint.

1 Introduction

Endogenous extracellular fields can affect spike timing and may coordinate activity across cortical regions (McFadden, 2002, 2020; Pockett, 2012; Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015). Coordination raises a further question: when do those regions describe one situation, and when does the resulting state represent its own place in it?

Consider reaching for a cup. Vision and movement can describe different features while agreeing on the cup’s location. Matching their timing leaves agreement about that shared quantity untested. An internal description of the perceiver’s relation to the cup asks for still more. We call the three requirements *coherence* (collective phase order), *compatibility* (agreement of local descriptions on overlaps), and *self-representation* (accurate reconstruction from an internal encoding). The mathematical states are spatial measures and fields; identifying them with perceptual content requires independent decoding evidence.

Binding-by-synchrony and communication-through-coherence accounts

17 treat phase relations as the medium of integration or of selective routing
 18 (Singer & Gray, 1995; Fries, 2015), and field theories place integration in the
 19 endogenous electromagnetic field (McFadden, 2020; Pockett, 2012). Both
 20 leave open what fact about synchronized activity makes it one content rather
 21 than coordinated activity carrying several. The contribution here is to sep-
 22 arate the three requirements and prove which follow from which. Phase
 23 order constrains shared content only through encoders related on overlaps,
 24 with a bound that grows as the square root of the population it is read
 25 on. Compatible local descriptions, not synchrony, determine a unique global
 26 state. A contracting self-map fixes one state but can reconstruct accurately
 27 only a family of bounded diameter. Each result names a measurement that
 28 phase data alone cannot supply.

29 The familiar noisy mean-field oscillator supplies a stationary coherence
 30 threshold (Sakaguchi, 1988; Strogatz & Mirollo, 1991). We add a resource
 31 condition for a declared actuator, bounds that connect phase order to con-
 32 tent only through specified encoders, and conditions under which compatible
 33 descriptions glue and an accurate reconstruction fixes the glued state. A
 34 Lean 4 development checks these implications and a joint witness satisfies
 35 their eight connecting hypotheses in a specified model. It proves neither that
 36 cortex satisfies those hypotheses nor that the glued state is experienced. The
 37 latter identification motivates the framework and needs its own endpoint.

38 The empirical question is which measurements could support or discon-
 39 firm the proposed connections. A phase resultant can miss a locally or-

40 dered winding field; stationary phase distributions can leave the coupling
 41 mechanism unidentified; and coherent adaptation can lose alignment with
 42 a designated content template. Independent decoders must identify shared
 43 quantities before their agreement can be tested. We therefore distinguish sta-
 44 tionary branches from finite observed onset, and compare prespecified field
 45 and connectivity onset models on held-out emergence recordings. The com-
 46 parison tests a specified spatial prediction, not the presence of experience.
 47 Section 10.2 states the comparison that bears on conscious content directly.
 48 Sections 2–5 develop the three requirements; Section 6 applies their mea-
 49 surement obligations to a digital candidate. Section 7 states the resource and
 50 composition premises, and Sections 8–9 examine observables and the cortical
 51 test. Table 1 lists the eight connecting hypotheses; Supplemental Material
 52 gives their derivations and controls.

53 **2 Limits of phase order**

54 Several limits hold without the cortical hypothesis. Equal phases can coexist
 55 with incompatible local descriptions; phase order constrains shared content
 56 only through encoders related on overlaps. A winding field can have zero
 57 global resultant while remaining locally ordered. A constant readout can
 58 have a unique fixed point without reconstructing its input. A region that
 59 does not distinguish relevant states cannot encode them, and a site cannot
 60 report a change outside its causal past by a declared deadline. These lim-

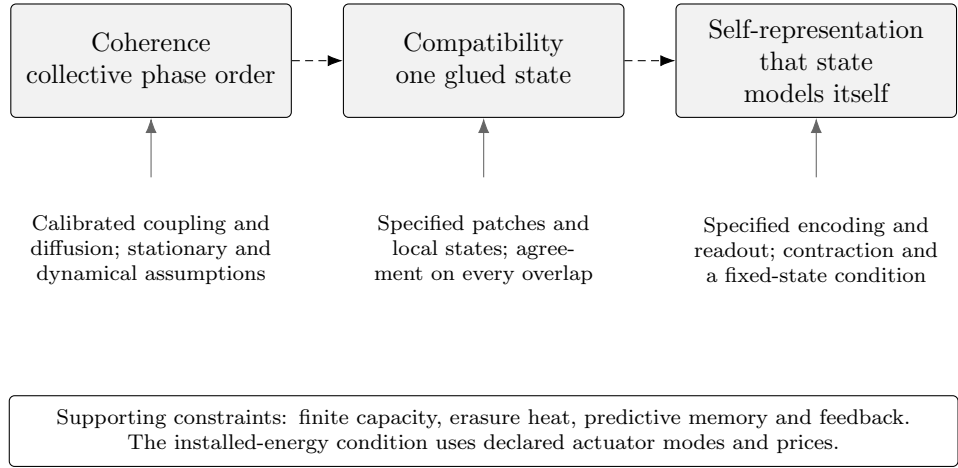


Figure 1: Three requirements of the proposed account. Each requires the inputs shown below it. Dashed arrows mark proposed connections: phase coherence alone supplies neither compatible local descriptions nor an accurate self-model. Thermodynamics constrains a candidate implementation through separate assumptions. This is the explanatory structure of the article; Table 1 records the eight implications consumed by each branch of the formal composition.

61 its reflect familiar phase oscillator and information constraints (Acebrón et
 62 al., 2005; Ott & Antonsen, 2008; Dörfler & Bullo, 2014; Breakspear, 2017).
 63 The sections below state each one beside the condition or measurement it
 64 motivates.

65 3 Coherence: physical coupling and its re- 66 source condition

67 3.1 Dynamics and the stationary threshold

68 The finite stochastic model is

$$d\theta_i = \left[\omega_i + \sum_j K_{ij}(t) \sin(\theta_j - \theta_i) \right] dt + \sqrt{2D} dW_i, \quad r_N = \left| \frac{1}{N} \sum_i e^{i\theta_i} \right|. \quad (1)$$

69 Here θ_i is the phase of oscillator i , ω_i its natural frequency, $D > 0$ is phase dif-
70 fusion, W_i are independent Wiener processes, and K_{ij} has inverse-time units,
71 with angles dimensionless. Uniform mean-field coupling is $K_{ij} = K/N$; fre-
72 quency heterogeneity and diffusion are distinct parameters. The stationary
73 calculation below uses identical frequencies and this scalar coupling. Ap-
74 plying it to a spatial kernel requires an aggregation rule and a check of the
75 resulting approximation.

76 For identical frequencies in their rotating frame, the positive normalized
77 stationary densities of the one-body Fokker–Planck equation are exactly

$$\rho(\theta) = \frac{e^{a \cos(\theta - \psi)}}{2\pi I_0(a)}, \quad a = \frac{Kr}{D}, \quad r = R(a) := \frac{I_1(a)}{I_0(a)}, \quad (2)$$

78 where I_n are modified Bessel functions and ψ is the collective phase. The for-
79 mal derivation uses the absence of stationary current for a periodic gradient
80 drift to obtain this density, then verifies normalization and self-consistency.

81 At $K \leq K_c = 2D$, only $r = 0$ is self-consistent (Sakaguchi, 1988; Strogatz &
82 Mirollo, 1991). Above threshold, exactly one positive solution coexists with
83 zero and increases strictly with K . Its onset is

$$\frac{r^2}{K - K_c} \rightarrow \frac{1}{D}, \quad r \sim \sqrt{\frac{K - K_c}{D}}. \quad (3)$$

84 Figure 2 shows this stationary branch.

85 Linearization about the uniform density gives rate $K/2 - D$ for the first
86 harmonic and $-Dn^2$ for harmonics $n \geq 2$. A growing mode therefore exists
87 exactly above threshold. About the coherent density, the quadratic form
88 is strictly dissipative transverse to collective rotation, whose rate is zero.
89 These linear estimates concern the limiting equation. Relating them to finite
90 stochastic trajectories requires a dynamical mean-field limit and branch selec-
91 tion; exchange symmetry alone does not establish the needed independence
92 (Sznitman, 1991).

93 **3.2 Installed energy bounds mean coupling**

94 A spatial model replaces the finite sum by integration against a kernel. One
95 declared implementation carries response modes with continuous profiles ϕ_i ,
96 installation prices u_i and nonnegative occupancies $c_i(\Xi)$ in configuration Ξ :

$$K_\Xi(x, y) = \sum_i c_i(\Xi) \phi_i(x) \phi_i(y), \quad U(\Xi) = \sum_i u_i c_i(\Xi). \quad (4)$$

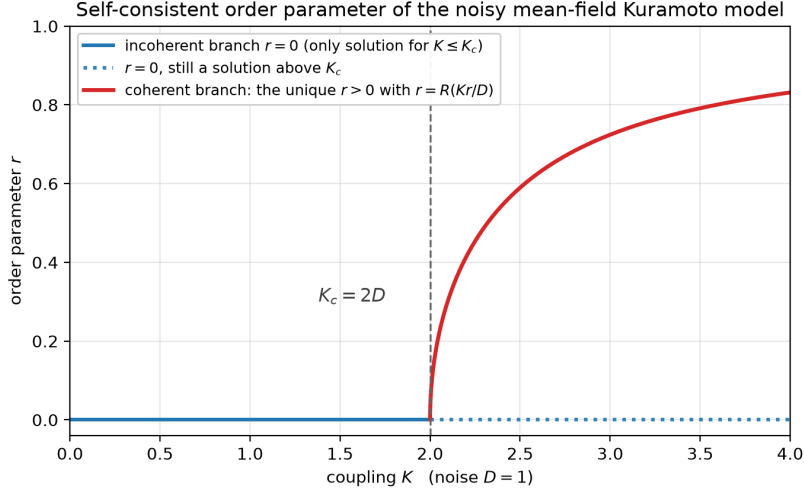


Figure 2: Stationary self-consistency for identical frequencies and phase diffusion $D = 1$. The positive coherent branch appears continuously above $K_c = 2D$ and increases with coupling. Zero remains a solution above threshold. The curve is computed from the integral expressions used in the formalization.

97 The same configuration sets the reciprocal kernel and stored energy. An
 98 executed transition changes the kernel through its changed occupancies and
 99 requires work $U(\Xi') - U(\Xi) + Q$, with Q positive into the reservoir. The
 100 installed kernel is averaged over the resulting configuration law $p(\Xi)$. Cell-
 101 pair reconstruction on refining spatial partitions converges to its product-
 102 measure integral under the specified compactness and sampling assumptions;
 103 the refinement index is independent of the number of control updates.

104 Normalize the substrate measure μ to one. Write $m_i = \int \phi_i d\mu$ and
 105 declare $\kappa > 0$ with $m_i^2 \leq \kappa u_i$ for every mode. Factorizing each mode's

106 integral gives

$$\begin{aligned}
 U_{\text{inst}} &= \sum_{\Xi} p(\Xi) U(\Xi), \\
 K_{\text{mf}} &= \iint K(x, y) d\mu(x) d\mu(y) = \sum_{\Xi} p(\Xi) \sum_i c_i(\Xi) m_i^2 \leq \kappa U_{\text{inst}}.
 \end{aligned} \tag{5}$$

107 If this mean coupling is the K of the scalar model, $\kappa U_{\text{inst}} \leq 2D$ excludes a
 108 coherent stationary branch and any growing Fourier mode about the uniform
 109 density. Thus coherence on that branch requires $U_{\text{inst}} > 2D/\kappa$. The condition
 110 is necessary: energy above it need not produce sufficient coupling because
 111 the bound can be loose.

112 The theorem concerns stored energy in the declared actuator modes. The
 113 estimated cortical signalling expenditure and extracellular electrostatic en-
 114 ergy in Supplemental Material measure different quantities; neither identifies
 115 the occupancies, mode prices or their relation to metabolic supply. Conse-
 116 quently the bound supplies a necessary condition for a declared actuator,
 117 but no cortical value of κ or demonstration that cortex meets it. Applying
 118 the scalar threshold to heterogeneous or signed spatial dynamics also needs
 119 a justified reduction.

120 4 Compatibility: from coordinated activity 121 to a global description

122 4.1 Why common timing is insufficient

123 For overlapping patches U and V , compatibility means that restricting either
124 local description to $U \cap V$ gives the same object. Descriptions can agree on
125 shared quantities while contributing different information elsewhere. Com-
126 patible sections of the specified sheaf have a unique global extension. On
127 a finite cover, local finite spatial measures agreeing as measures on every
128 overlap also have a unique finite-measure extension whose sheaf image is the
129 glued section. No mass-one normalization is imposed.

130 In the cup example, visual and reaching descriptions must express a
131 shared quantity, such as cup location, in common coordinates. Restriction
132 retains that quantity while dropping details particular to each description.
133 Agreement permits surface appearance and hand configuration to contribute
134 to one situation. The mathematical counterexample assigns the same phase
135 to every patch while two local mass profiles disagree on a shared site. Those
136 profiles have no common extension. Because they are chosen independently
137 of phase, synchrony places no constraint on their disagreement.

138 This distinction limits the biological interpretation. Receptive territories
139 and content domains must be related to spatial patches, and the descriptions
140 must support the required extension. Matching overlap marginals of de-

141 coded probability laws need not determine a unique global joint distribution
 142 (Abramsky & Brandenburger, 2011).

143 The cover is itself a choice, and the construction constrains it in one way
 144 only: synchronized patches agree where they overlap. Where they do not
 145 overlap there is nothing to compare, so a cover by pairwise disjoint patches
 146 satisfies that obligation for *every* assignment of local descriptions and phases
 147 and the gluing theorem still returns a unique global section, which is the
 148 tuple it was handed. Membership of the class then reports nothing about the
 149 system beyond the patches covering it. What a gluing claim asks of a cover
 150 is therefore a non-degeneracy the theorem does not assume: that distinct
 151 patches meet, and that what they share is a quantity fixed independently of
 152 the agreement it is being used to establish.

153 4.2 When coherence does constrain content

154 A declared relation between phase and content gives a positive result. Let
 155 contents take values in a real normed vector space $\mathcal{C}$. Patch i reads an
 156 assigned oscillator through $e_{i,x}$ at site x , evaluated on the circle point
 157 $(\cos \theta, \sin \theta)$. Assume a uniform Lipschitz constant L in chord distance and
 158 require $e_{i,x} = e_{j,x}$ wherever patches i and j overlap. Encoders may vary across
 159 sites; their equality on overlaps is a supplied relation between descriptions of
 160 the shared quantity. Write $z_k = e^{i\theta_k}$ and $m = N^{-1} \sum_k z_k$, so $|m| = r$. The
 161 phasors' squared deviations from their mean sum to $\sum_k |z_k - m|^2 = N(1 - r^2)$,
 162 and $|z_i - z_j| \leq |z_i - m| + |z_j - m|$ gives $|z_i - z_j|^2 \leq 2N(1 - r^2)$ for each pair.

163 Consequently every overlap discrepancy is bounded by

$$\delta_r = L\sqrt{2N(1-r^2)}. \quad (6)$$

164 Here discrepancy is the norm of the difference, and r and N are read on
165 whichever population the bound is applied to. That choice is what decides
166 whether the estimate says anything. A patch carries an order parameter
167 of its own, and reading the bound there carries that patch's resultant and
168 that patch's site count, so a large population observed through small patches
169 is constrained by the small count. Read instead on the whole array, the
170 factor $\sqrt{N}$ makes the estimate weak, because it draws a pointwise guarantee
171 from an average over N sites. The order cannot be improved in general:
172 one oscillator in antiphase among N leaves $N(1-r^2)$ near 4. At $r = 1$ the
173 observations agree exactly at either scale. A shared scalar encoder is a special
174 case. Dropping the overlap relation permits disagreement even at perfect
175 locking. A common nonexpansive linear projection preserves agreement, but
176 agreement of a projection does not establish agreement of the full vector.

177 What Eq. (6) bounds is a chord, which never exceeds 2, so the estimate
178 constrains anything only where the right-hand side falls below that. At the
179 sheet's own patch order, the bound is informative for at most 10 sites; beyond
180 that, the coherence-to-content link requires the bridge assumption E78 of
181 the cortical hypothesis (Table 1 gives the full composition and its edges).
182 Supplemental Material provides the detailed derivation of this saturation,

183 and states three conditional derivations that replace the uniform factor $\sqrt{N}$
 184 by a count of disagreeing pairs, a chain across a cover's shared sites, or a
 185 spectral gap at a frequency-locked configuration.

186 Mix retained fields with observations at gain $\eta \in [0, 1]$: $s_i \mapsto (1 - \eta)s_i +$
 187 ηo_i . Content and observation discrepancies bounded by ε and δ give a next
 188 discrepancy at most $(1 - \eta)\varepsilon + \eta\delta$. Repeated observations bounded by δ
 189 therefore give

$$\varepsilon_n \leq (1 - \eta)^n \varepsilon_0 + \delta. \quad (7)$$

190 Thus agreeing observations preserve exact compatibility and, for $\eta > 0$, re-
 191 duce an initial mismatch. Approximate observations leave a nondecaying
 192 term in the error bound. Taking a uniform bound on Eq. (6) connects this
 193 update to the phase process.

194 4.3 What approximate agreement determines

195 Exact gluing requires exact restrictions. For nonnegative profiles on a finite
 196 substrate, suppose every overlap discrepancy is at most ε in uniform distance.
 197 Weights that vanish outside their patches and sum to one at each site select
 198 a global profile by weighted averaging. The selection lies within ε of each
 199 patch there, and any two such selections differ by at most ε , independently
 200 of cover multiplicity. An L -Lipschitz readout consequently varies by at most
 201 $L\varepsilon$ across selections.

202 At zero tolerance, the selection is the unique exact extension. At every

203 positive tolerance there are families with no exact extension; selection then
 204 gives an approximation whose dependence on weights is bounded. Supple-
 205 mental Material gives a three-site witness attaining the bound. The mass-
 206 profile construction uses the scalar case of the content dynamics and addi-
 207 tionally requires nonnegativity of the initial profiles and observations. The
 208 general vector result supplies no identification with the spatial-measure sheaf.

209 These constructions concern declared objects: a specified cover, contents
 210 valued in a normed space, and encoders related on overlaps. They select
 211 neither the neural cover nor the decoder, and identifying their contents with
 212 perceptual content, or establishing that the identification bears on experi-
 213 ence, is a further modelling task; Section 8 sets out what a measurement of
 214 it would have to establish.

215 **5 Self-representation: a state that models it-** 216 **self**

217 **5.1 The reflexive construction**

218 Compatibility gives a global state. Reflexivity relates it to an encoding within
 219 that state. Equip global sections with a complete metric. If the additional
 220 restriction-resonance condition identifies encoding with restriction to a region

221 $\mathcal{A}$, the reconstruction map is

$$F(s) = \text{readout}(s|_{\mathcal{A}}). \quad (8)$$

222 The region and readout are fixed across the state space. If F contracts and
 223 fixes the section s_* formed from the local descriptions, Banach’s theorem
 224 makes s_* its unique fixed point (Banach, 1922). The formal bridge E89
 225 (Table 1 gives the full composition and its edges) supplies the contraction
 226 and $F(s_*) = s_*$; restriction resonance is separate. Supercritical coupling
 227 makes the stipulated contraction factor $\Lambda = e^{-(K-2D)\tau/2}$ less than one for
 228 a relaxation time $\tau > 0$. Constructing a map with that factor remains a
 229 modelling obligation. A fixed contraction can reconstruct accurately only a
 230 restricted family of states: if its error is at most ε throughout that family,
 231 any two members are at most $2\varepsilon/(1 - \Lambda)$ apart.

232 In the cup example, the global state includes the represented scene and the
 233 perceiver’s relation to it. A local encoding must retain enough information
 234 for the readout to recover that modelled state; cup position alone would be
 235 insufficient if hand configuration and viewpoint could not be reconstructed.

236 For the same map with Lipschitz factor $\Lambda < 1$, approximate reconstruc-
 237 tion gives

$$d(s, F(s)) \leq \varepsilon \implies d(s, s_*) \leq \frac{\varepsilon}{1 - \Lambda}. \quad (9)$$

238 At the stipulated rate this radius grows as $2\varepsilon/((K-2D)\tau)$ near threshold, for
 239 fixed positive ε and τ . Exact uniqueness survives while the error guarantee

240 weakens.

241 5.2 Reconstruction across a family of states

242 Sensitivity can be required quantitatively. Declare relevant macrostates, an
243 encoder Φ and readout R , and require $d(s, R(\Phi(s))) \leq \varepsilon$ throughout that
244 family. If two states have the same code, the triangle inequality gives

$$\Phi(s) = \Phi(t) \implies \max\{d(s, R(\Phi(s))), d(t, R(\Phi(t)))\} \geq \frac{1}{2}d(s, t). \quad (10)$$

245 Thus M relevant states pairwise more than 2ε apart require at least M
246 distinguishable codes; an alphabet of at most 2^b codes requires $M \leq 2^b$.
247 Applied to the reflexive boundary, E is its local encoding and $R \circ E$ is the
248 existing self-prediction map.

249 This is an additional accuracy criterion on a family, with no requirement
250 for multiple exact fixed points. For the same fixed contracting map used by
251 E89, however, the diameter bound above is an additional restriction: at zero
252 error the family contains at most one metric-distinct state. A broad family
253 therefore requires a different, possibly input-conditioned reconstruction map
254 with its own accuracy and stability assumptions. A blind encoder or constant
255 readout fails the stated criterion for separated states despite having a unique
256 fixed point.

257 Where the encoding is read as the state's own restriction to a region, the
258 criterion becomes a condition on the region. Restriction resonance requires

259 the encoder and that restriction to identify the same pairs of states, so a
 260 boundary reconstructing two relevant states more than 2ε apart while its
 261 region reads them identically is not resonant. Read forwards, a resonant
 262 boundary that reconstructs a pairwise separated family restricts injectively
 263 on it. Which states a region reads identically is fixed by where the region
 264 sits, before an encoding is chosen, so the resolution the family demands is
 265 the region's and no readout supplies it. The implication runs one way: a
 266 region that does resolve the family is not thereby shown to carry a resonant
 267 encoding, and E89 supplies no such resolution.

268 Resonance and resolution are jointly satisfiable without a local fold. Take
 269 the region to be the whole substrate and use the identity readout: restric-
 270 tion is the identity, so this boundary is resonant and reconstructs every de-
 271 clared family exactly. Its map has Lipschitz factor one and does not satisfy
 272 E89's strict contraction premise on a nontrivial metric space. A properness
 273 condition on the region is therefore a separate requirement, and the fixed-
 274 contraction diameter bound remains in force when E89 is also required. With
 275 an L -Lipschitz readout, a family reconstructed within ε whose members lie
 276 pairwise more than $d_{\min}$ apart has δ -separated codes whenever $\delta L \leq d_{\min} - 2\varepsilon$,
 277 so the family is no larger than the δ -packing number of what the region car-
 278 ries. That is the shape a continuous family of states needs, and at the whole
 279 substrate it says nothing, because the region's range is everything. The gain
 280 L is declared and derived here for no physical readout, and at unbounded
 281 gain the bound says nothing.

282 **5.3 Agreement after a change takes communication**

283 Snapshot compatibility does not specify how quickly descriptions respond
284 when a shared quantity changes. In a synchronous network whose updates
285 read only the site’s own state and declared neighbours’ messages, two initial
286 configurations agreeing on a site’s causal past through T rounds produce the
287 same state there at round T . The deadline consequence is the one Section 2
288 states, and it follows from the reconstruction bound in Section 5.2: a report
289 from that site is a readout of a code that the two interventions leave identical.

290 A three-site delay line witnesses failure after one round and exact report-
291 ing after two; cutting the path prevents delivery at every round. A report
292 may still happen to be correct about one fixed world. The bound concerns
293 responsiveness across interventions.

294 Throughout this section *prediction* names a reconstruction map, and the
295 relevant family of states, the metric, the tolerance, the communication graph
296 and the deadline are all declared inputs, as are the represented variables and
297 the fidelity counted as accurate. Reading the map as temporal neural pre-
298 diction would require dynamics; reading the resulting code count as physical
299 memory capacity would require a hardware model, which a state-dependent
300 readout would have to extend.

301 6 What the conditions constrain in a digital 302 candidate

303 The conditions bear on a large language model (LLM) running on graphics
304 processing units (GPUs) only through measurements, and only for a declared
305 task. For a finite spatial coupling operator with eigenvalues $|\lambda_1| \geq \dots \geq |\lambda_N|$,
306 any linear encoder and readout factoring through d real coordinates pays re-
307 construction error at least $\sum_{i>d} \lambda_i^2$. That rank premise is a property of an
308 explicitly linear bottleneck; query/key dimension does not supply it for soft-
309 max attention. On one synthetic two-view task whose target is the glued
310 state and a self-relative coordinate, a rank-two linear decoder has predicted
311 training-error floor 2.000 and held-out error 2.000 on a parity-held-out family.
312 The deadline premise is similarly specific: within one forward pass, the causal
313 mask confines position t to its causal past, so an intervention outside it can-
314 not be reconstructed there by that step. These results rank no architecture
315 family, imply no general inferiority of GPU hardware, and settle no question
316 of machine experience; a claim of unity or self-representation for such a sys-
317 tem commits its maker to identifying the shared quantities, the code and its
318 separation of the declared family under intervention. Supplemental Material
319 gives the bottleneck derivation, the task controls, the information ceiling of
320 token outputs and the hardware scope.

321 7 Resource premises and composition

322 Two thermodynamic statements enter, with different standing. The first is
323 proved for the phase model: for a smooth positive periodic density with prob-
324 ability current J and diffusion D , the current functional $\sigma_J = \int J^2/(D\rho) d\theta$
325 bounds how fast order can change, $\int_0^\tau \sigma_J dt \geq [\arcsin r(\tau) - \arcsin r(0)]^2/(D\tau)$
326 while $0 < r < 1$, and stationary order has no positive floor. The second is
327 conditional: the installed-energy condition of Section 3.2 is necessary for its
328 declared actuator, and restricts a physical candidate only once mode prices,
329 coupling reduction and diffusion are measured. Neither is a heat or power
330 statement until the current is identified with a physical flux, a reservoir at a
331 stated temperature is specified and local detailed balance holds for the steps
332 concerned; only then does σ_J convert to entropy production and the finite
333 register budgets to Landauer heat (Landauer, 1961; Norton, 2005; Shenker,
334 2000). No composition edge derives coherence, compatibility or reconstruc-
335 tion from a heat bound. Finite-register capacity obeys $H(p) \leq \log |X|$, and
336 a separate boundary result concerns a field reaching disconnected vacuum
337 components (Kibble, 1976); Table 1 states the connecting assumptions. Sup-
338 plemental Material gives the current-cost derivation, predictive-memory and
339 feedback balances, learning and repeated-operation budgets, and the sensi-
340 tivity of the metabolic comparison to its assumptions.

341 7.1 What the composition establishes

342 The passive and feedback composition theorems each accept eight connecting
343 implications (Table 1). In the active branch, a named substep at the register’s
344 thermal scale satisfies

$$k_B T > 0, \quad k_B T [H(p) - H(p')] \leq \langle Q \rangle \leq B, \quad (11)$$

345 with B the register’s heat. The upper allocation follows from a ledger whose
346 operations deliver that total heat to one reservoir and whose other opera-
347 tions do not increase joint entropy. Those ledger assumptions are physical
348 inputs. Neither Landauer’s lower bound nor microscopic reversibility sup-
349 plies them. Finite two-bit witnesses exhibit positive forward and reverse
350 elementary paths, local detailed balance at one thermal scale, and interac-
351 tion energies of order $k_B T$; these conditions alone do not quantify proximity
352 to equilibrium. Cortex has not been shown to satisfy these premises. The
353 illustrative cortical metabolic expenditure over one region and one phase-
354 diffusion time is 2.4×10^{14} times kT at body temperature, but identifying it
355 with the register heat budget B requires a common physical ledger. Read-
356 ing the register at the scale where its premises are physical, such as a single
357 channel, requires E45Active to carry that step’s budget to the convergence
358 of a continuum kernel. The active branch is an alternative route to the same
359 node rather than a stronger one, and no result of Section 3.2 passes through
360 it: the installed-energy condition is derived from E56 and E67.

Table 1: Connecting hypotheses, eight per branch, and the installed-energy consequence. The branches share E12, E23 and E56–E89. Exact overlap compatibility and the encoding–readout structure are additional inputs. Supplemental Material, Table S1, maps the node-level results.

Connection	Status	Required assumption
E12	Physical	A finite-capacity system realizes a field
<i>capacity and</i>	premise	leaving its vacuum manifold. Capacity and
<i>boundary</i>		boundary are independent roots.
E23	Bridge	The boundary is associated with a finite
<i>boundary to</i>	assumption	register whose update leaves states
<i>register</i>		unreachable.
E34 /	Bridge	Passive: predictive memory at the register’s
E34Active	assumption	thermal scale and heat budget, with
<i>register to</i>		nonzero waste. Active: positive paths, local
<i>dissipation</i>		detailed balance and allocation of register
		heat to the named substep; the allocation
		uses the ledger identity and other
		operations’ compressiveness. Both are
		exhibited at thermal-scale heats on finite
		state spaces with positive reverse paths,
		which is not the regime a cortical register
		occupies.
E45 /	Modelling	Convergent spatial coupling energies.
E45Active	assumption	Declared mode hardware supplies a kernel
<i>couplings to</i>		limit for either branch. A learning witness
<i>continuum</i>		computes installation and work from its
		own executed law. Convergence follows
		from mesh refinement rather than from the
		budget, which selects no regime at
		macroscopic scale. Refinement and physical
		identifications remain inputs.
E56	Physical	A specified physical field realizes the
<i>field</i>	commitment	limiting coupling and satisfies the kernel
<i>identification</i>		conditions.
E67	Physical	That field operates above the
<i>above</i>	commitment	synchronization threshold.
<i>threshold</i>		

Table 1 (continued)

Connection	Status	Required assumption
E78 <i>coherence to cover</i>	Bridge assumption	Coherence relaxes the supplied cover at coupling at least the named mean-field value. Shared phase encoders bound vector-content mismatch; the exact cover and timely responsiveness remain separate requirements. Patches and oscillators carry one index, so one oscillator carries exactly one content patch: a modelling assumption rather than a consequence, and not an innocent one, since field-mediated coupling is spatial while two territories describing one quantity need not be neighbours. Where the cover's nerve is connected the same edge carries through the sites its patches share, at a residual linear in that nerve's diameter; where the configuration is frequency-locked it carries through the coupling graph's algebraic connectivity λ_2 instead of through any patch count, which asks the cover for nothing and is declared hardware data in the standing of κ .
E89 <i>contraction to fixed point</i>	Modelling assumption	A map mediated by local self-encoding fixes the glued section and has the stipulated Lipschitz factor $\Lambda < 1$. Reconstruction over a declared family imposes an additional code-separation criterion.
Installed energy	Conditional theorem	Under Section 3.2's occupancy and mode-price conditions, E56 and E67 require $U_{\text{inst}} > 2D/\kappa$. The learning witness clears this necessary bound; its matched control does not. Neither physical premise follows.

361 Both branches also take a thermodynamic cover carrying exact over-
362 lap compatibility and a reflexive boundary specifying encoding and read-
363 out. They conclude that the glued section is the unique fixed point of self-

364 prediction. Compatibility and accuracy at that section enter as premises;
365 uniqueness follows. Identifying encoding with restriction needs the separate
366 resonance condition of Eq. (8).

367 Each of the eight edges is supplied rather than derived from the one above
368 it. E12 is equivalent to its conclusion, so capacity and boundary are inde-
369 pendent roots rather than a step from one to the other; both E45 discharges
370 take their convergence from refinement data and not from the thermody-
371 namic bound; E78 states that coherence relaxes the supplied cover, and the
372 coherence-to-content estimate of Eq. (6) bounds disagreement only for en-
373 coders already shared on overlaps. The inferential content of the composition
374 is its last step, where contraction and a fixed glued section give uniqueness.
375 What composing them establishes is that the premises are consistent, named
376 in one place and machine-checked together, and not that coherence follows
377 from installed energy or self-representation from coherence.

378 Joint witnesses combine a register, boundary, refining mesh and small
379 spatial cover; active witnesses add controlled thermal updates and a priced
380 installation. They establish satisfiability under their stated identifications.
381 Mesh convergence follows from refinement data, and the witnesses establish
382 no common cortical mechanism. Supplemental Material maps the claims to
383 their Lean declarations and records the counterexamples. The development
384 declares no additional axioms beyond its libraries' foundational axioms.

385 8 What the observations establish

386 The composition holds under its premises. Whether observation can estab-
387 lish those premises is a separate question, and the three requirements it
388 connects concern different objects: phase dynamics, local descriptions and
389 internal encodings. Their observables must retain these distinctions. The
390 following controls show which mechanisms remain indistinguishable, which
391 temporal approximations fail, which states one observable cannot separate,
392 and what independent content measurements must establish. Each is a finite
393 numerical regime on a declared model, with full parameters and controls in
394 Supplemental Material, and none of them calibrates a cortical quantity.

395 8.1 Stationary distributions leave the mechanism un- 396 determined

397 Equation (2) gives a concentration–coherence relation with no fitted shape
398 parameter. Estimate r from circular resultant length and a from density
399 shape: inverting R from the measured r would make agreement automatic.
400 Even independently estimated agreement identifies a statistical family shared
401 by different mechanisms.

402 Let $q(t)$ cross 2, let $r(q)$ be its stationary branch, and fix $D_0 > 0$. Increas-
403 ing coupling $K(t) = D_0 q(t)$ at diffusion D_0 and decreasing diffusion $D(t) =$
404 $K_0/q(t)$ at coupling K_0 both give concentration $a = q(t)r(q(t))$. Uncoupled
405 oscillators with prescribed drift $-h(t)\sin(\theta - \psi)$, where $h(t) = D_0 q(t)r(q(t))$,

406 and diffusion D_0 have that same density. Their stationary distributions co-
 407 incide window by window despite different causes.

408 Resolved dynamics supply additional information. In the ideal model
 409 with continuous latent paths, quadratic variation accumulates at rate $2D$
 410 and conditional increment means constrain drift; a filtered observation of
 411 the phase needs its own observation model before either is read off. Separat-
 412 ing endogenous coupling from a matched common drive additionally requires
 413 a measured drive or controlled perturbation; identifying electromagnetic cou-
 414 pling requires evidence specific to that interaction. Supplemental Material
 415 gives the equivalence construction and its numerical check.

416 **8.2 Finite ramps change the observable onset**

417 The square-root law in Eq. (3) describes stationary solutions. A changing
 418 coupling requires time for the phase distribution to approach those solutions.
 419 In a deterministic mean-field control that crosses the threshold symmetrically,
 420 none of the scanned rates meets the declared tracking tolerance: slowing
 421 the crossing does not remove the terminal residual under that initialization.
 422 A slow control variable therefore supplies no general guarantee of branch
 423 tracking at onset. Supplemental Material gives the residual, rate range and
 424 admissible rates away from threshold.

425 The finite-population ramp study uses $N = 2000$, $D = 1$ and 32 replicas
 426 at 8 coupling speeds from $v = 0.1$ to $v = 0.0001$. Escape is the first of three
 427 consecutive saved samples at $r \geq 0.2$ after threshold. Faster ramps delay

428 escape (Fig. 3); only 5/32 replicas escape within the fastest window. The
 429 6 uncensored legs supply the delay scaling below; the decade-spaced ones
 430 among them have mean coupling delays 0.1947, 0.0684 and 0.0253. Fitted
 431 onset exponents differ from those of the exact stationary branch evaluated
 432 on the same grids. At the fastest speed the ensemble mean never reaches the
 433 fit window, so its fitted slope does not estimate stationary onset.

434 Delayed passage through a noisy bifurcation (Berglund & Gentz, 2002)
 435 makes the apparent exponent depend on ramp speed and fitting window.
 436 The specified deviation-based critical ramp speed is not bracketed by this
 437 sweep. With fluctuating coupling, the relevant stationary comparison also
 438 depends on rate: the mean coupling approximates order in the tested fast
 439 regime, while averaging stationary order over coupling works for the tested
 440 slow drives confined to one side of threshold. Slow crossing drives can satisfy
 441 neither approximation.

442 **The prediction.** Over the 6 uncensored legs a finite-range fit gives $\Delta K \propto$
 443 $v^{0.444}$. The interval from fitting those means is [0.394, 0.494]; resampling
 444 replicas independently within each speed gives a 95% interval [0.398, 0.496],
 445 with fraction 0.019 of 10000 bootstrap fits at or above 1/2. This fraction
 446 describes the resampling distribution under the fitted protocol; it is not a
 447 test of the ideal exponent. A deterministic mean-field run seeded at the
 448 fluctuation floor gives exponent 0.425 with the same estimator and escape
 449 criterion; tightening its criterion to $r \geq 0.05$ gives 0.490. These controls

450 show that the criterion affects the fitted exponent, but do not prove that it
 451 accounts for the entire finite-population shortfall. The larger-population low-
 452 threshold fit has substantial log residuals; matched-speed fits do not converge
 453 toward one-half across the tested sizes and criteria. Some trajectories also
 454 cross the absolute escape level before K_c , so the first postcritical crossing can
 455 yield a zero or grid-limited delay. This finite sweep establishes no common
 456 asymptotic exponent.

457 The ideal delayed-onset calculation motivates a square-root scaling hy-
 458 pothesis; its exponent is generic to suitable bifurcations and does not iden-
 459 tify the coupling medium. A within-subject emergence design can vary the
 460 propofol taper and test how an operational onset changes with that manipu-
 461 lation, but interpreting the result as a coupling-rate law requires a measured
 462 or calibrated relation between taper and $\dot{K}$. The escape criterion must be
 463 set relative to each recording's pre-threshold fluctuation scale. Section 8.1
 464 describes how D could be estimated from resolved phase traces.

465 Additional controls address integration bias, population size and spatial
 466 or signed coupling; the refined signed-network crossing bracket $K_{\text{eff}}/D \in$
 467 $[1.958, 2.009]$ contains 2.

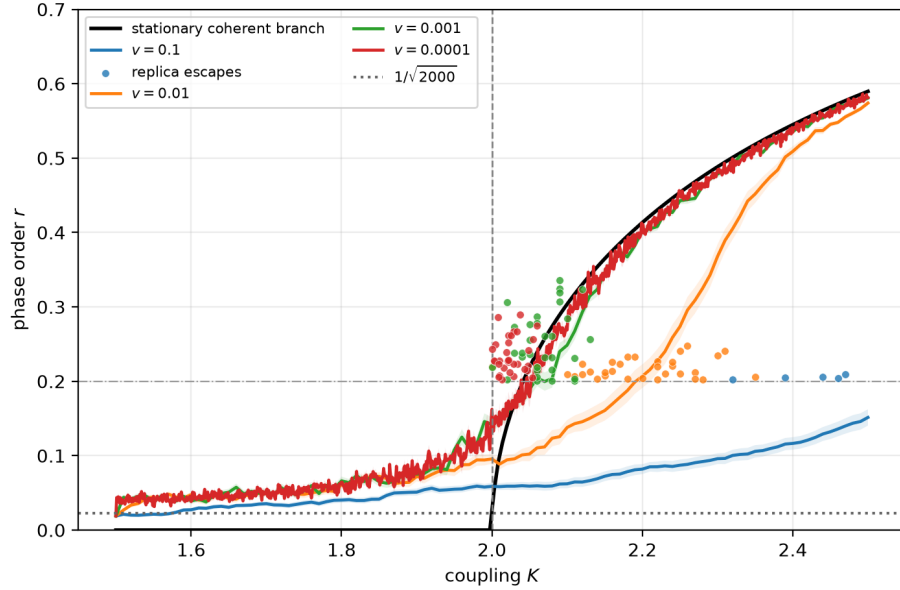


Figure 3: Finite-rate departures from the stationary branch. Curves show ensemble-mean order for four ramp speeds v , with one-standard-error bands. Coloured points mark each replica’s first sustained escape, plotted at its recorded order; censored replicas have no point. The dashed vertical line is the static threshold, the dash-dotted horizontal line is the escape criterion, and the dotted horizontal line is the finite-size reference. The replica-bootstrap interval for the delay exponent is reported in the text.

468 8.3 A winding state leaves the order parameter am- 469 biguous

470 The global order parameter r discards spatial arrangement. On a periodic
471 sheet, a phase field can wind once around the sheet and have $r = 0$ while
472 neighbouring sites remain locked. At identical frequencies, every such wind-
473 ing is stationary for an isotropic symmetric kernel: contributions at opposite
474 separations cancel. It violates the quarter-turn initial-spread premise of the

475 convergence result, which is why that result can still select the uniform state.

476 In the spatial control, the winding persists below the interpolated de-
 477 cay length 0.1871 mm and unwinds above it (Fig. 4). At 0.1 mm, uniform
 478 and winding starts under the same coupling, diffusion and kernel end at
 479 $r = 0.9657$ and $r = 0.0083$; patch-local order is 0.9015. Under quenched fre-
 480 quency spread 1.5 rad s^{-1} , the measured retention boundary is 0.1421 mm,
 481 and the corresponding order comparison is 0.9418 against 0.0348 over 3 frozen
 482 realizations. These are finite controls, not a cortical wave identification or a
 483 derived transition.

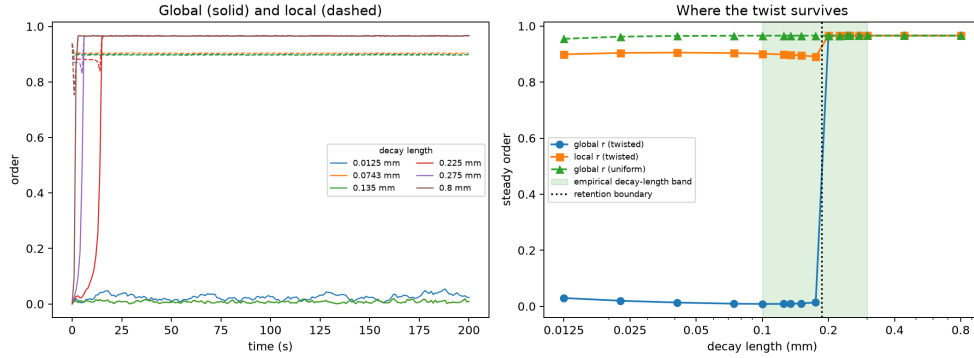


Figure 4: A winding phase field on the spatial sheet, at identical frequencies. Left: global order (solid) against patch-local order (dashed) across a subset of coupling decay lengths. Right: the twisted and uniform initial conditions compared across lengths, with the shaded 0.1–0.3 mm band and the interpolated retention boundary. Below that boundary the two initial conditions reach different stationary values of r at the same coupling, diffusion and kernel, so the resultant does not report the coupling. The sweep is sampled at the lengths marked; it establishes no transition and no biological identification.

484 Linearization about a winding reweights each separation’s coupling by

485 the cosine of its phase advance. The resulting symmetric kernel has explicit
 486 mode rates: nonnegative reweighted coupling gives no growing mode, while
 487 a positive rate destabilizes that winding. The criterion concerns linear rates,
 488 not asymptotic stability, and the measured retention boundary need not
 489 equal its limit. An amplitude extension adds an existence condition: $\Gamma + \lambda >$
 490 0 , where Γ is linear gain and $\lambda = \sum_d f_d(\cos \Delta\theta_d - 1)$ comes from the same
 491 kernel. For the nonnegative kernels used here, a broad kernel can remove a
 492 winding from that model even though the phase-only equations retain it as
 493 a stationary state.

494 Neither model turns these states into measured cortical spirals. The pe-
 495 riodic sheet cannot carry a lone phase defect, the relaxational dynamics has
 496 no propagation speed, and the amplitude variable has no calibrated relation
 497 to a recorded envelope. Supplemental Material gives the spectral, amplitude
 498 and defect results. Locally, Eq. (6) can still bound content disagreement on a
 499 winding patch when shared encoders are supplied, although its global $r = 0$
 500 form is uninformative. A nonzero ring winding number q also obstructs a
 501 global real-valued phase lift; it has no identified content meaning here. Thus
 502 an onset criterion based only on r can register a change of basin as well
 503 as a change of coupling. The protocol of Section 9.3 needs a spatial phase
 504 observable.

505 8.4 Coherent adaptation can lose environmental align- 506 ment

507 An adaptive coupling rule might supply representational structure, but its
508 objective must reward the relevant agreement. For symmetric coupling, pair-
509 wise sine terms cancel in the population mean drift. Minimizing squared drift
510 at that fixed mean therefore rewards equalizing drift across nodes. In the
511 frequency-cluster control, this favors coupling between unlike-frequency clus-
512 ters, whereas the diagnostic environmental template rewards coupling within
513 clusters.

514 The formalization proves descent of the squared-drift objective for a spec-
515 ified gradient flow at fixed phases. Its functional differs from thermodynamic
516 entropy production: at a coherent stationary density the probability current
517 vanishes while phase-averaged squared drift is positive. Supplemental Mate-
518 rial gives the operator, its gradient and the conditions of the descent result.

519 The finite control co-evolves noisy phases and symmetric, nonnegative
520 couplings at fixed total resource. Its template is a diagnostic covariance
521 of latent signals, never a training input. Across three illustrative seeds,
522 second-half order remains between 0.9167 and 0.9222, while the within-
523 cluster to between-cluster coupling ratio falls from 0.9824–0.9948 to 0.3507–
524 0.6135 (Fig. 5). Random-update, permuted-gradient and frozen-kernel con-
525 trols are reported in Supplemental Material.

526 The immediate post-update objective reduction disappears when aver-

527 aged over complete update intervals. A prespecified grid of 6 rate–cadence
528 combinations contains no case meeting the 5% sustained-reduction criterion
529 while retaining coherence and losing alignment. The finding is coherent adap-
530 tation away from the specified structure, with no demonstrated sustained cost
531 reduction.

532 Kernel alignment and overlap compatibility are different observables.
533 This control addresses acquisition of its designated template under its par-
534 ticular objective. Learning models can explicitly include data in an objective
535 (Corberi et al., 2026); an account of agreement about the cup must specify
536 the content its learning rule is meant to preserve.

537 **8.5 Content agreement needs independent decoders**

538 A neural test needs independently identified domains, local descriptions and
539 restriction maps. Two decoders should estimate a shared content on over-
540 lapping domains before their agreement is examined. Discrepancy requires a
541 null under decoding noise and controls using deliberately compatible and in-
542 compatible examples. An implementation on synthetic data separates these
543 cases at $AUC = 0.986$, on a 95% interval of $[0.980, 0.992]$ resampled by trial;
544 the compatible construction scores 0.476 on an interval containing one half.
545 Shared-prior shrinkage can nevertheless manufacture agreement, and regional
546 bias can manufacture disagreement. Per-patch decoding diagnostics detect
547 these failures; the compatibility statistic alone does not.

548 Phase agreement cannot replace that measurement. On constructed con-

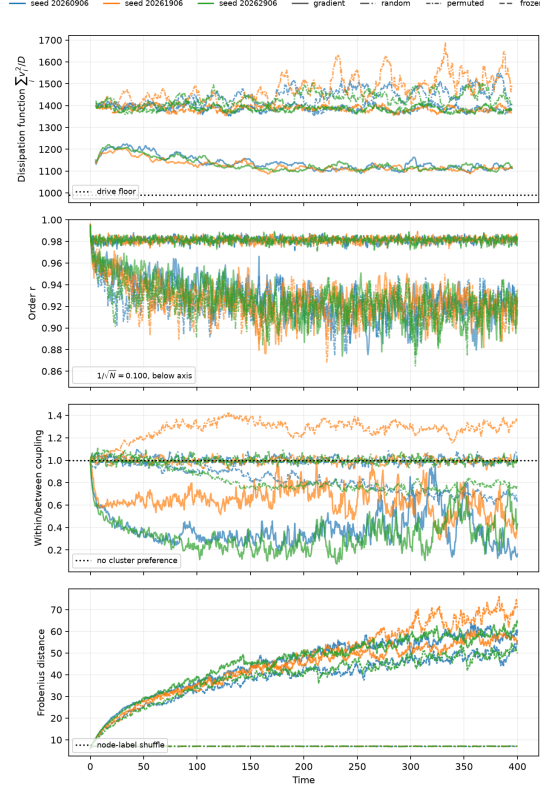


Figure 5: Coherence persists while coupling loses alignment with the frequency-cluster template. Gradient updates (solid), random updates (dash-dotted), permuted gradient updates (dash-dot-dotted), and a frozen kernel (dashed) share one phase-noise stream per seed. Panels show the immediate post-update squared-drift objective, phase order, within-cluster to between-cluster coupling, and template distances. The gradient arm preserves order while its coupling ratio falls; growth in both distances mainly reflects kernel norm. The objective panel samples the instant after updates; complete-interval measurements find no sustained reduction in the bounded study.

549 tent, phase carries a share 1.000 of the content entropy when content is built
550 from it and 0.000 when it is not. A phase-derived statistic reports com-
551 patibility on phase-locked patches whose local descriptions disagree. The
552 implemented content statistic detects their incompatibility. These controls
553 validate a measurement on known constructions; a neural content model and
554 decoder remain to be established.

555 Choosing the cover also chooses a candidate subject: trivial descriptions
556 can agree while carrying little content, and a partition can conceal disagree-
557 ment. The useful choice must therefore be constrained by what neural ac-
558 tivity encodes. The exact composition assumes compatibility of its supplied
559 cover. The content mechanisms above identify additional hypotheses under
560 which agreement can arise; they leave the empirical choice of content and
561 the connection to that cover explicit.

562 **8.6 Estimator calibration on exploratory EEG**

563 The concentration range and the dependence structure decide whether the
564 concentration-coherence relation can be tested at all. Under the specified
565 log-density estimator and bias-floor criterion, discrimination at $a = 0.542$
566 requires 3000 independent sites in the scanned designs, rising to 31000 when
567 sites share a mean in groups of 10. These counts describe that estimator and
568 dependence model at a declared design concentration; the intended recording
569 requires its own calibration. Supplemental Material reports the bin-count
570 controls and the full dependence-aware design.

571 The estimators were also exercised on public propofol-sedation EEG (Ba-
 572 jwa et al., 2025): 61 adjacent bipolar pairs formed from 62 scalp channels,
 573 band-passed, Hilbert-transformed and pooled over pairs and 100-ms win-
 574 dows. Across 8 subjects the pooled concentration spans 0.087–0.142, and
 575 the residual from the Bessel relation is small. That agreement is uninforma-
 576 tive. Pooling over a window does not remove the common rotation of the
 577 phases, so a perfectly phase-locked 10-Hz signal passed through the same
 578 pipeline returns $r = 0.0001$ and $a = 0.0004$ with zero residual, although its
 579 instantaneous spatial resultant is one. The bipolar montage also removes
 580 spatial coherence: a field whose raw channels have instantaneous resultant
 581 0.89 gives 0.06 across bipolar pairs and 0.007 pooled. Any nearly uniform
 582 pooled distribution satisfies $r \approx a/2$, and over the observed range the Bessel
 583 ratio departs from $a/2$ by at most 0.0002. The exercise therefore checks
 584 the estimator’s behaviour and establishes no spatial concentration, coher-
 585 ence level or test of the stationary density. Sedation recordings also leave the
 586 awakening hypothesis untested.

587 An informative dataset therefore needs an estimand that removes the
 588 common rotation, a montage whose effect on spatial phase is modelled, suf-
 589 ficient concentration range or precision, calibrated estimators and an ob-
 590 servation model for dependence and source mixing. Intracranial recordings
 591 (ECoG, sEEG or LFP) or narrowband burst analysis may reach a more
 592 informative concentration range, but that range must be measured. The
 593 design counts above are calculations for the specified estimator, not sam-

594 ple requirements transferable to another recording. Public mouse calcium
595 imaging provides thousands of spatially located neurons across waking, sleep
596 and anesthesia (Oomoto et al., 2026), but its 7.65 Hz frame rate is below the
597 phase-estimation band used here. Site count, state coverage and bandwidth
598 are separate requirements.

599 **9 The cortical hypothesis and an awakening** 600 **test**

601 The cortical proposal identifies a physical source of coupling and asks whether
602 its changes help explain recovery. The tests in Section 8 determine what
603 observations must distinguish before a phase trace can support this identifi-
604 cation.

605 **9.1 Physical identification and calibration**

606 The cortical hypothesis identifies part of effective coupling with endogenous
607 extracellular electromagnetic interactions. Fields can modulate spike timing
608 (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015), and field models
609 propose coordination of distributed neural information (Pinotsis & Miller,
610 2022; McFadden, 2020). Phase organization under propofol motivates study-
611 ing this regime (Bastos et al., 2021), but does not measure coupling.

612 The formal field predicate requires a jointly continuous kernel, a normal-

613 ized substrate, coupling equal to the kernel average, and no single site pro-
614 viding more than half that strength. Toy constructions satisfy it. A cortical
615 realization must specify the scale at which the kernel and its reduction ap-
616 ply: calcium imaging reports different state-dependent modularity patterns
617 at single-cell and mesoscale resolution (Kiyooka et al., 2026).

618 Cortical columns are the obvious candidate unit, and they answer one
619 of the two demands the formalism makes. As the oscillator index they are
620 a serious proposal: they are delineated by response properties and anatomy
621 rather than by the agreement they would be used to establish (Mountcastle,
622 1997), and a phase can be estimated from a column-sized territory at the
623 electrode spacings used for cortical wave measurements (Townsend et al.,
624 2015). As the content cover they are a poor one, for three separate reasons.
625 A parcellation into non-overlapping columns is a disjoint cover, which satisfies
626 the compatibility obligation without constraining anything (Section 4). The
627 cover condition additionally requires the patches to exhaust the substrate,
628 while much of association cortex has no accepted columnar parcellation. And
629 the column’s standing as a canonical cortical unit is contested (Horton &
630 Adams, 2005; Rakic, 2008). Offering columns for both roles at once therefore
631 supplies the oscillator index; the content cover remains an open empirical
632 question, which the bridge assumption E78 records.

633 The spatial reduction is also testable. A row-sum aggregation at zero
634 diagonal tracks the uniform-coupling control over the tested spatial decay
635 range, while frequency heterogeneity displaces the operational threshold. The

636 fit window biases both estimates relative to $2D$; agreement with the control
637 is therefore the supported conclusion. Supplemental Material reports the
638 reduction, its numerical error and the distinct assumptions needed for the
639 kernel-average identification in Eq. (5).

640 The field commitment requires physical evidence beyond a choice of math-
641 ematical representation, and the available support comparisons do not supply
642 it. A finite matrix spread over positive-mass cells preserves its total cou-
643 pling and phase correlation exactly, while point-supported vanishing under
644 an atomless measure and a disadvantage from excluding a better-correlated
645 pair concern specified support constraints; Section 6 reads the same compar-
646 isons against a digital candidate. Neither supplies energy prices.

647 A spike-time shift alone supplies no coupling rate. The dimensionless
648 product of population size N , timing sensitivity $\mathcal{T}$, field amplitude E and
649 frequency f needs a conversion rate γ :

$$K_{\text{eff}} = \gamma N \mathcal{T} E f. \quad (12)$$

650 This rate describes how induced timing shifts accumulate into drift. Nei-
651 ther it nor the installation factor κ follows from a heat bound. Testing the
652 physical threshold requires independent estimates of the interaction kernel,
653 its aggregation and diffusion in the same cortical regime.

654 9.2 Extracellular geometry and recovery

655 The proposed control variable is extracellular geometry. Extracellular vol-
656 ume fraction is approximately $\alpha = 0.2$ at tortuosity $\lambda \approx 1.6$ (Syková &
657 Nicholson, 2008); a sleep study reports a rise from roughly 0.14 to 0.23 (Xie
658 et al., 2013). Below 10 kHz, cortical tissue conducts approximately resistively
659 at $0.3\text{--}0.6\text{ S m}^{-1}$ (Barbour, 2017). At fixed transmembrane current density,
660 an effective-medium description predicts that a larger extracellular fraction
661 raises conductivity and reduces field amplitude.

662 The coupling-gated hypothesis consequently predicts that extracellular
663 contraction on awakening raises E and, holding the other factors in Eq. (12)
664 fixed, raises effective coupling. This argument supplies a direction; the
665 geometry–coupling relation, its magnitude and its role in human recovery
666 require measurement.

667 For a smooth crossing at nonzero speed, evaluating the stationary branch
668 along the changing control gives a deterministic tracking approximation that
669 fails at onset in the tested crossing control:

$$r_{\text{ss}}(K(t)) \sim \sqrt{\frac{\dot{K}(t_0)}{D}} (t - t_0)^{1/2}, \quad K(t_0) = K_c, \quad \dot{K}(t_0) > 0. \quad (13)$$

670 This is a deterministic tracking approximation from Eq. (3) at fixed D ; the
671 crossing controls in Section 8.2 show that it fails as an observed-onset ap-
672 proximation under the tested initialization. A separation of geometry and
673 phase time scales can support tracking away from threshold. The awakening

674 hypothesis needs a calibrated dynamical prediction for an operational onset
675 above the finite-size fluctuation floor. Crossing K_c alone does not supply a
676 recovery latency.

677 **9.3 A protocol for the awakening experiment**

678 Neural inertia describes a behavioural barrier and hysteresis between anaes-
679 thetic induction and emergence (Friedman et al., 2010); recovery can pass
680 through metastable activity states (Hudson et al., 2014; Proekt & Hudson,
681 2018). The $v^{1/2}$ exponent of an ideal delayed bifurcation is generic (Baer
682 et al., 1989; Berglund & Gentz, 2002) and cannot alone discriminate a field
683 mechanism from a synaptic one. A possible discriminator is spatial onset
684 relative to independently estimated local field coupling and network connec-
685 tivity. The present mean-field model has no spatially varying κ and derives
686 no nucleation order, so a field-based onset map needs a separate model spec-
687 ified before the observations are inspected.

688 For that comparison, partition the recording into prespecified regions i
689 and measure pre-onset order $r_i(t_0)$ and diffusion D_i . Define a field score
690 $h_i = N_i \mathcal{T}_i E_i f_i$ from independent pre-onset field and timing measurements,
691 and a connectivity score c_i from a separately estimated structural network.
692 For the same measured emergence drive $g(t)$, compare $K_i^F(t) = a_F g(t) h_i$ with
693 $K_i^C(t) = a_C g(t) c_i$. Each model has one fitted scale. The local first-harmonic

694 approximation predicts

$$\widehat{r}_i^M(t) = r_i(t_0) \exp\left(\int_{t_0}^t [K_i^M(s)/2 - D_i] ds\right), \quad M \in \{F, C\}. \quad (14)$$

695 Predicted onset is the first crossing of a region-specific threshold fixed from
696 pre-onset fluctuations. Fit the two scales on training sessions, then com-
697 pare held-out onset times and site order under the same observation model.
698 Larger field scores predict earlier onset only when baseline order and diffu-
699 sion do not reverse the ranking. This declared approximation omits interre-
700 gional coupling and saturation; a successful ranking would support its field
701 covariate, not prove ephaptic causation. Supplemental Material specifies the
702 controls and failure conditions.

703 A within-subject protocol can manipulate emergence rate, with spatially
704 resolved recordings to compare the proposed field and connectivity models:

- 705 1. **Manipulate emergence rate.** Vary the propofol infusion taper
706 across sessions within the same subject, recording drug concentrations
707 or a validated pharmacokinetic proxy to quantify the manipulation.
- 708 2. **Measure behavioural recovery.** Record time-to-response to a stan-
709 dard stimulus as the independent behavioural endpoint, rather than
710 estimating ΔK directly.
- 711 3. **Resolve spatial onset.** Identify anatomical locations where coher-
712 ence first crosses the baseline-scaled threshold and score the two maps

713 of Eq. (14) on held-out sessions. The two covariate maps are fixed
714 before the outcome sessions are inspected.

715 **4. Match physical controls.** Confirm that the observed coherence-to-
716 amplitude relation holds under the specific local diffusion D and ob-
717 servation noise. The model’s quadratic-variation rate $2D$ applies to
718 the latent phase; a band-limited analytic phase is smooth below its
719 correlation time, and on a constructed field its squared-increment rate
720 changes 37-fold between 0.2-ms and 20-ms steps. Estimating D there-
721 fore requires an explicit observation model for filtering and sampling.

722 Predeclare the recording window, regional fluctuation thresholds, and a
723 shared timing-error model. Fit the two scales and that observation model
724 on training sessions only; on held-out sessions, score the observed site-onset
725 times and sites that never cross by the same regional predictive rule. Report
726 site order as a secondary outcome and behavioural time-to-response sepa-
727 rately. Sessions without a crossing are censored observations, not zero delays
728 or exclusions. Compare the two scores only where independently estimated
729 field and connectivity maps predict distinguishable onset patterns at the
730 recording resolution. A held-out field-model advantage would support that
731 specified spatial predictor; it would not identify an ephaptic cause, shared
732 content or experience. Supplemental Material gives the controls and pilot
733 quantities needed to judge whether the comparison is informative.

734 **10 Discussion**

735 **10.1 Unity and a minimal self as interpretive motiva-** 736 **tion**

737 We propose the state glued from compatible local descriptions as a formal
738 correlate of conscious unity, and its accurate internal reconstruction as a cor-
739 relate of a minimal self in the sense of a pre-reflective point of view on the
740 scene (Gallagher, 2000). In the cup example, this would mean one description
741 that preserves the cup’s relation to the perceiver as well as the scene. Com-
742 patibility permits distinct contents and uncertainty; the shared quantities
743 must be chosen from independent evidence about what each region encodes.
744 The reconstruction need not be a verbal or autobiographical account. Its
745 proposed relation to experienced perspective remains an interpretation. Nei-
746 ther identification is among the eight hypotheses in Table 1, and satisfying
747 them would not measure experience.

748 **10.2 A comparison that could disconfirm the link to** 749 **content**

750 The framework predicts that conscious content tracks compatibility, not co-
751 herence. Present matched stimuli that are reported on some trials and not
752 on others, for example at masking or rivalry threshold, and fit decoders for a
753 shared stimulus quantity independently on two overlapping cortical regions.

754 Compare decoded agreement on the overlap between reported and unreported
755 trials after matching their phase coherence. The proposed link predicts higher
756 agreement on reported trials at equal coherence. Equal agreement, or agree-
757 ment that follows coherence and not report, counts against it. A self-model
758 comparison has the same form: reconstruction of perceiver-relative variables,
759 such as hand position relative to the target, from a declared local encoding
760 should be more accurate on reported trials. These are correlational tests of
761 the proposed correlates; report is itself an access measure, and a no-report
762 design would be needed to separate content from access.

763 **10.3 What a test of the full proposal must establish**

764 A test needs separate estimates of effective coupling and diffusion, shared
765 content on overlapping regions, and reconstruction from a specified encoding
766 at a declared fidelity. The cover, relevant state family and decoder must
767 be fixed without selecting them for agreement. An independent experiential
768 endpoint is needed to evaluate the interpretation; absence of a report alone
769 does not define absence of experience. A unified episode with incompati-
770 ble decoded descriptions would challenge the unity identification. A unified
771 episode lacking the proposed self-encoding would challenge the minimal-self
772 claim. Conversely, satisfying the mathematical conditions without such an
773 episode would challenge their sufficiency as correlates.

774 For the physical route, cortical actuator modes and prices, the geometry-
775 coupling relation and a dynamical onset model remain to be measured. The

776 field/connectivity comparison in Section 9.3 evaluates one spatial onset pre-
777 diction. Controlled changes in coordination and independently decoded con-
778 tent agreement would address the representational connection. The synthetic
779 decoder exercise calibrates part of these measurements, and the EEG exercise
780 identifies what a pooled bipolar estimand cannot measure; neither identifies
781 a neural content code or tests experience.

782 10.4 Explanatory scope

783 Integrated-information, global-workspace and predictive-processing accounts
784 organize unity, access and modelling differently (Tononi et al., 2016; Baars,
785 1997; Dehaene et al., 2014; Friston, 2010). This account specifies conditional
786 relations among coupling, local content and reconstruction. Its formal field
787 condition requires a continuous kernel, a normalized substrate, a specified
788 average coupling and noise, and a site-contribution bound. Those conditions
789 do not identify an electromagnetic substrate; the extracellular interpretation
790 needs the separate calibration of Section 9. The joint witness shows that
791 the formal hypotheses can hold together in a declared model. A cortical
792 realization and an experiential interpretation require additional evidence.

793 Three limitations bound every claim above. Synchrony is not itself an
794 experiential measure, and nothing here establishes a consciousness criterion:
795 the conditions are proposed correlates to be tested against report or another
796 independent endpoint. The present EEG analysis checks a measurement
797 pipeline and does not validate any experiential claim. The formal results are

798 conditional on declared encoders, covers and maps whose cortical counter-
799 parts are unmeasured.

800 The remaining explanatory question is whether these relations occur in
801 neural activity and matter for experience. The framework identifies candidate
802 correlates and tests of their physical and representational premises. It does
803 not explain why satisfying them would be accompanied by experience.

804 11 Conclusions

805 Phase coherence, compatibility and self-reconstruction are separable condi-
806 tions. Coherence constrains shared content only through encoders shared
807 on overlaps and with a bound that grows as the square root of the popula-
808 tion read; compatible local descriptions, not synchrony, determine one global
809 state; and a contracting self-map fixes that state while reconstructing only a
810 family of bounded diameter. Each result replaces an assumption about syn-
811 chrony by a measurement: independently decoded agreement on overlaps,
812 a declared encoding and its fidelity, and calibrated coupling and diffusion.
813 The comparison of Section 10.2 states how such measurements could count
814 against the proposed link to conscious content.

815 Notation

Table 2: Symbols used throughout, with the section that defines each. Where one letter carries more than one meaning, the meanings are separated by the section they are read in.

Symbol	Defined	Meaning
θ_i, ω_i	Eq. (1)	Phase and natural frequency of oscillator i
N	Eq. (1)	Number of oscillators in the population the quantity is read on
D	Eq. (1), Supp.	Phase diffusion in inverse time; in the Supplemental Material’s convergence proof, the phase spread $D \leq \pi/2$
K_{ij}, K	Eq. (1)	Pairwise and scalar mean-field coupling, $K_{ij} = K/N$
K_c	Eq. (2)	Synchronization threshold, $K_c = 2D$
r, r_N	Eq. (1)	Resultant length (order parameter), on the population it is read on
a, ψ	Eq. (2)	Concentration $a = Kr/D$ and collective phase of the stationary density
q	Section 8.3	Winding number of a phase field read around a ring
μ	Eq. (4)	Substrate measure, normalized to one
Ξ, ϕ_i, u_i, c_i	Eq. (4)	Actuator configuration; mode profile, installation price and occupancy of mode i
U_{inst}, κ	Eq. (5)	Installed energy and the mode-data conversion factor bounding $K_{\text{mf}} \leq \kappa U_{\text{inst}}$
L	Section 4.2	Lipschitz constant of an encoder or readout
ε, δ	Section 4.2	Content and observation discrepancy tolerances
M, b	Eq. (10)	Size of a declared family of states and code length in bits
Λ, τ	Eq. (8)	Contraction factor $\Lambda = e^{-(K-2D)\tau/2}$ and its relaxation time
T	§7, §5	Temperature in the thermal scale $k_B T$; number of communication rounds to a deadline
ν	Supplemental Material	Fan-in of a communication graph: the most sites any one site hears from
$k_B T, B$	Eq. (11)	Thermal scale, read as one symbol, and heat budget of a register

Table 2 (continued)

Symbol	Defined	Meaning
$\gamma, \mathcal{T}, f$	Eq. (12)	Field-to-coupling conversion rate, timing sensitivity and frequency
α	Section 9	Extracellular volume fraction
λ	§8.3, §9	Eigenvalue the coupling kernel supplies at a winding; extracellular tortuosity
Ω, λ_2	Section 4.2	Common rotation rate of a frequency-locked configuration, and the algebraic connectivity of the coupling graph
β, k	Section 4.2	Chord diameter of a patch and the diameter of a cover's nerve, in hops
$\mathcal{C}, e_{i,x}$	Section 4.2	Content space and the phase encoder patch i reads at site x
Φ, R	Eq. (10)	Encoder and readout of the reconstruction criterion
$\mathcal{A}, F$	Eq. (8)	Encoding region and the reconstruction map $F(s) = \text{readout}(s _{\mathcal{A}})$
z_i, Γ, A_{ij}	Section 8.3	Complex amplitude, linear gain and coupling matrix of the amplitude layer
$\Delta\theta_d$	Section 8.3	Phase a winding advances across separation d
E	Eq. (12)	Field amplitude, in the cortical identification
U	§3.2	Stored energy $U(\Xi)$ of a configuration
s, s_*	§4–5	A global section and the one the reconstruction fixes

816 Data and code availability

817 The Lean development, simulation scripts, tests and saved numerical sum-
818 maries are maintained in the GitHub repository `bobaseb/coupling_gated_`
819 `cortical_coherence`. The formalization uses Lean 4.34.0-rc2 and Math-
820 lib commit `5ce203ecd53f8c3e47ee5530d7c86b94a55f0183`. Publication

821 simulation values are generated from saved summaries without rerunning
822 production sweeps. The exploratory EEG analysis uses OpenNeuro dataset
823 ds005620 (Bajwa et al., 2025) (subjects 1010, 1016, 1022, 1033, 1045,
824 1054, 1060, 1067, resting awake and sedation runs 1–2); preprocessing
825 and selection are described in Supplemental Material and implemented in
826 `simulations/empirical_collapse.py`. The repository contains the repro-
827 ducible analysis and does not redistribute the raw recordings.

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831 **Use of AI tools**

832 AI assistance to this work was substantial and is not separable by task.
833 Several coding assistants and chat interfaces were used in an interleaved way
834 throughout: Google’s Antigravity CLI (Gemini 3.1 Pro, Claude 4.6 Opus,
835 Claude 4.6 Sonnet), OpenAI’s Codex CLI (GPT-6 Astra and Sol; GPT-
836 5.6 Terra, Luna and Sol), Claude Code (Claude Opus 5 and Opus 5.5),
837 Gemini Pro through its web interface, and Anthropic’s Fable and DeepSeek
838 v4 Flash through OpenRouter, accessed with the Hermes CLI. Between them
839 they contributed to the Lean development, the simulation and analysis code,

840 and the drafting and revision of both documents. Because the tools were
841 used in combination and in alternation, no section, proof, figure or result is
842 attributable to any one of them, and none is claimed to be. The author is
843 responsible for all content, including everything a tool produced. Machine
844 checking establishes the stated Lean results under their hypotheses; it does
845 not validate biological interpretations or replace scientific review of generated
846 code and prose.

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